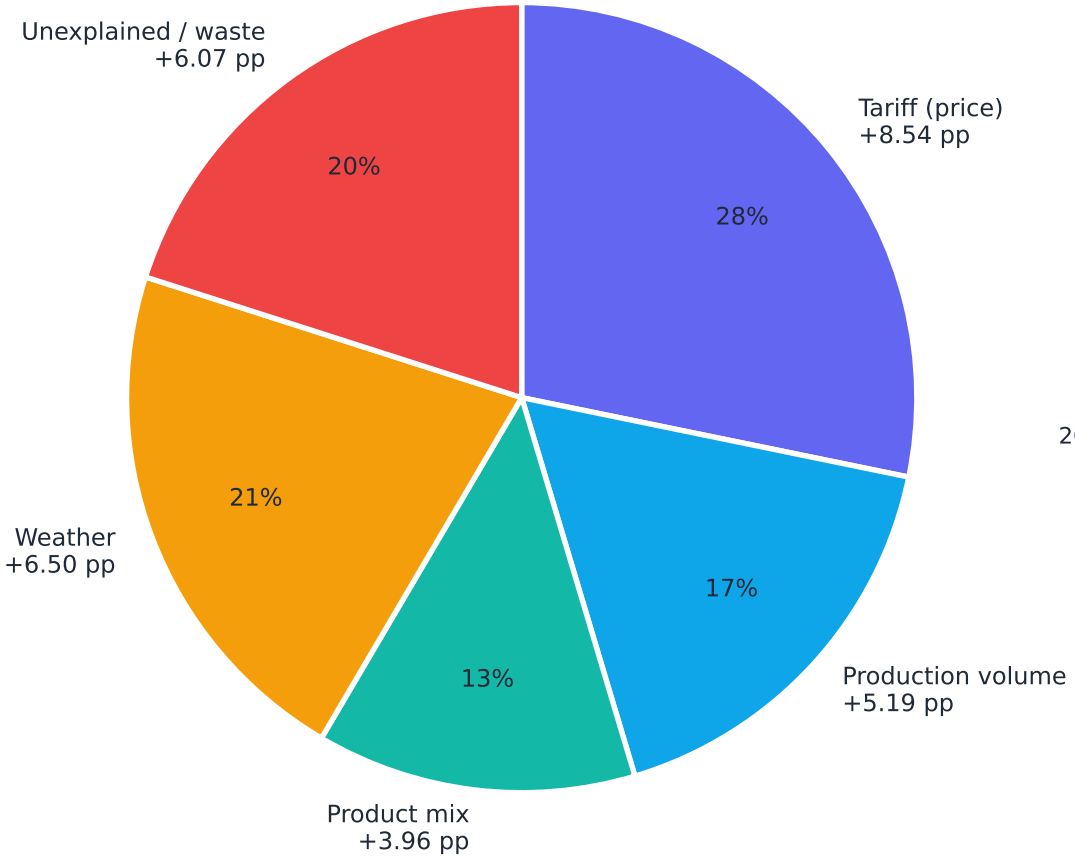
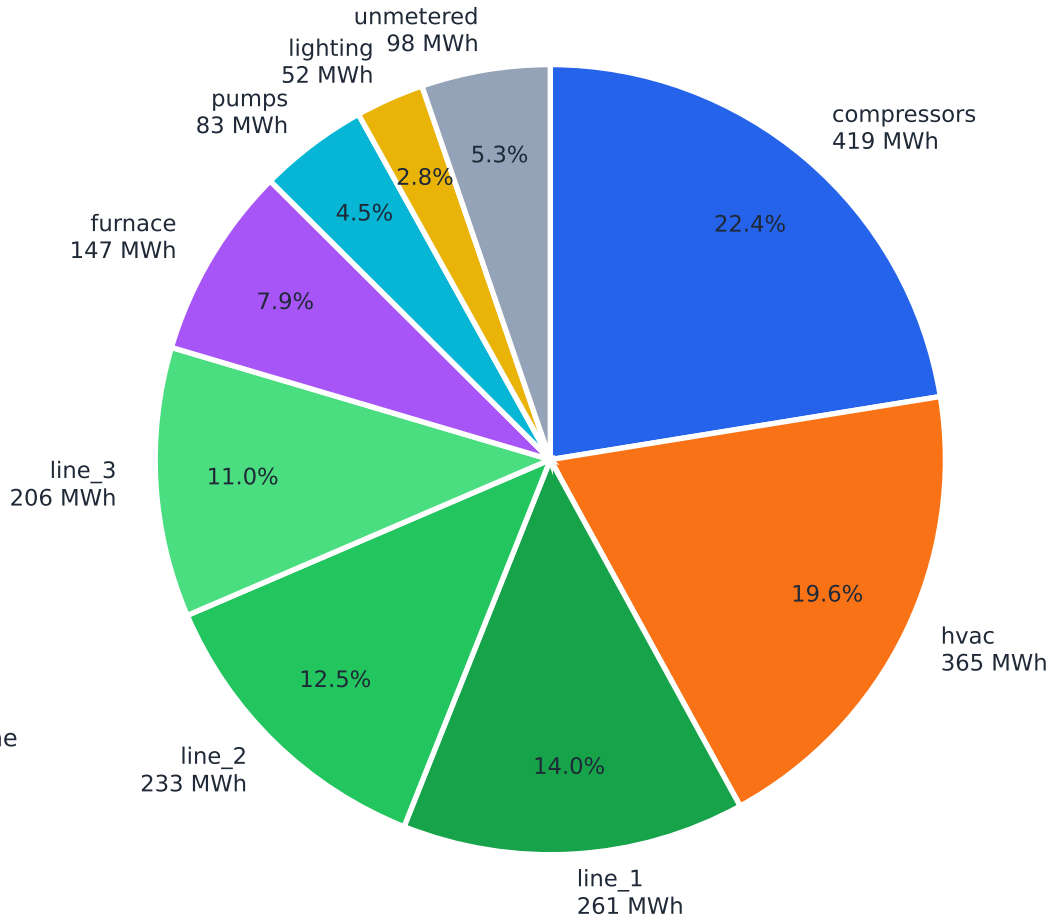
Where the money and the energy went: 2026-Q2 to 2026-Q3

Computed by the app from data/history. Bill 249,597 -> 325,113 (+30.26%). Baseline learned from 2025-Q3, 2025-Q4, 2026-Q1, 2026-Q2.

Bill increase of 30.26% split into 5 causes
(share of the increase)



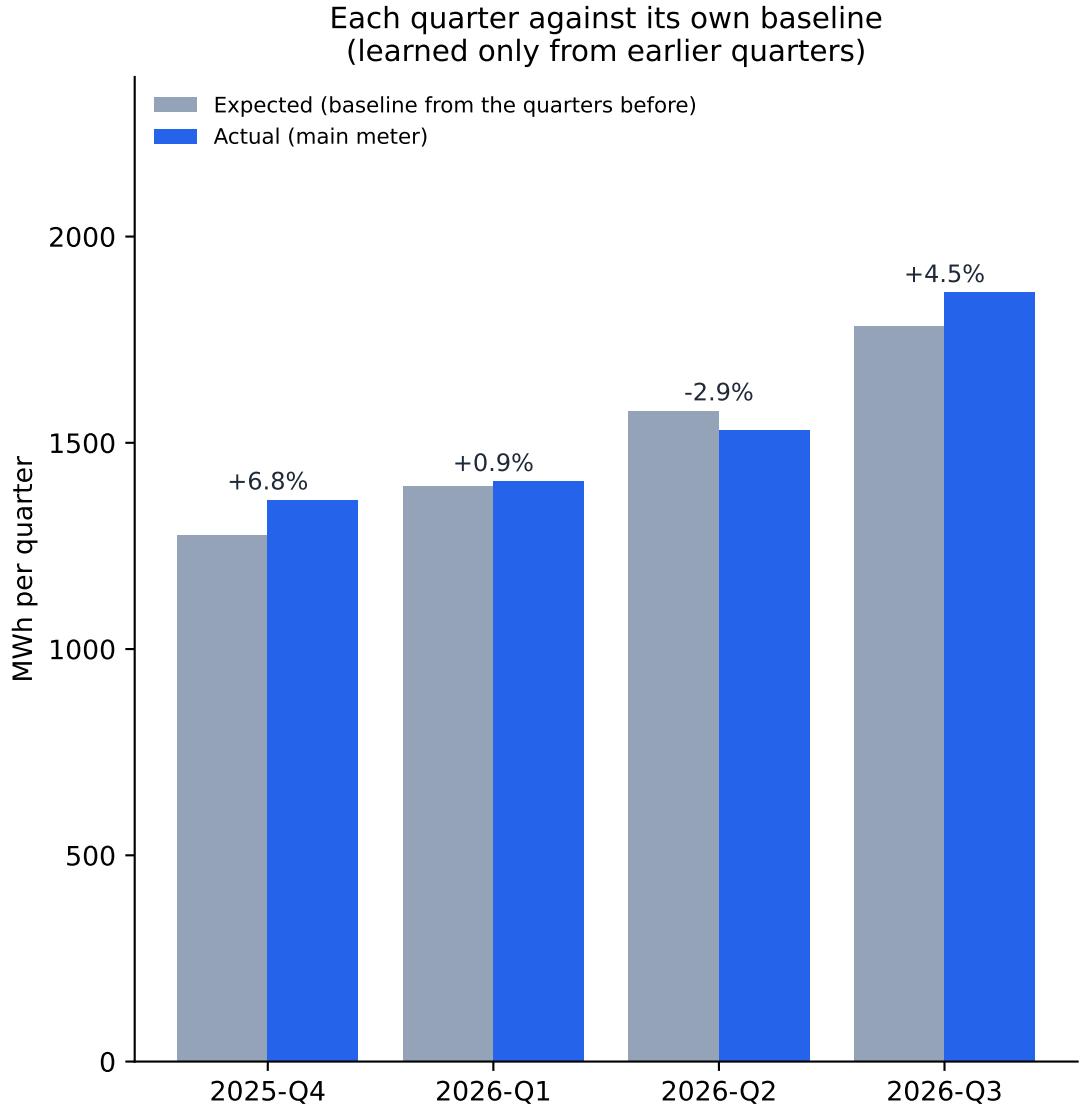
2026-Q3 energy by sub-meter
(main meter 1,865 MWh)

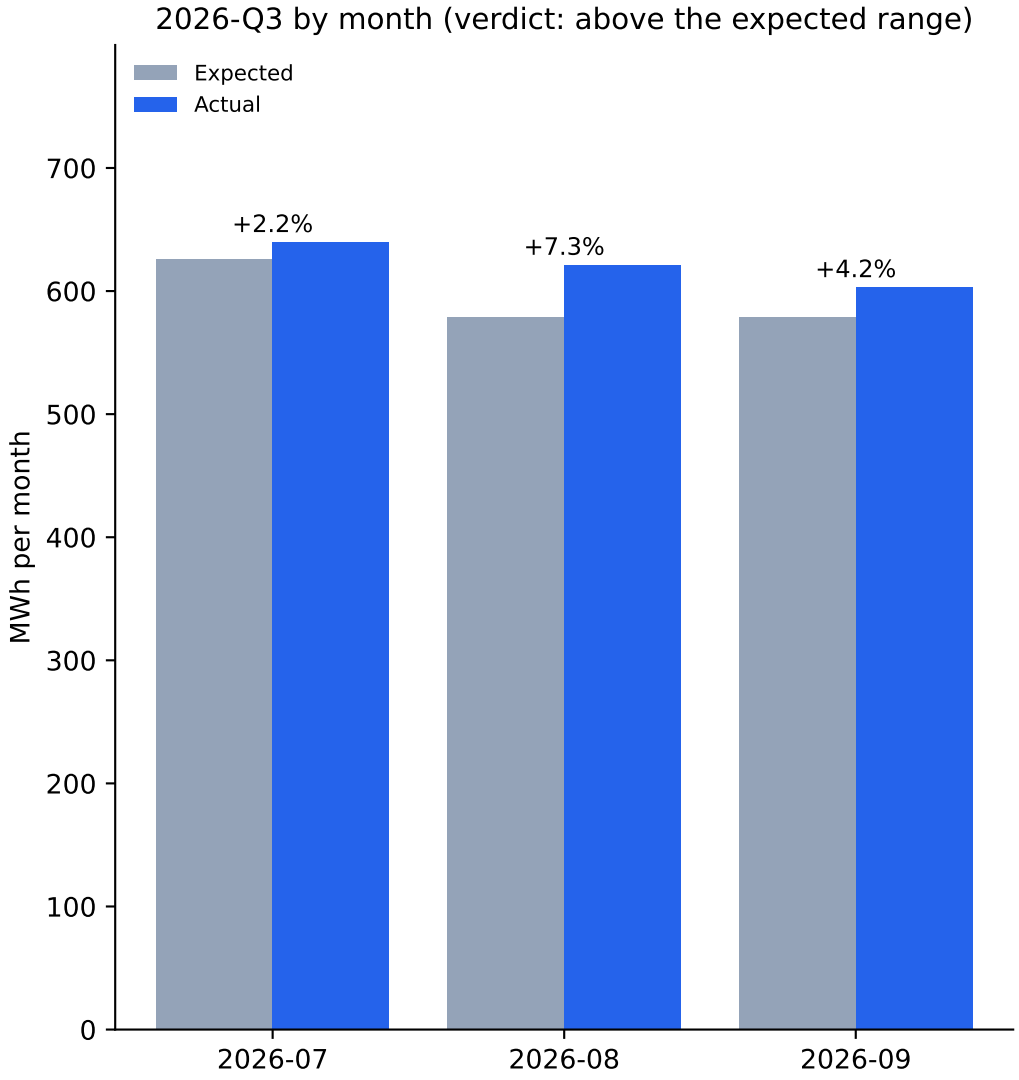


pp = percentage points of the previous quarter's bill. The five slices add up exactly to the total change. Unmetered = main meter minus the sum of sub-meters.

The baseline: expected energy vs actual energy

Linear regression of daily kWh on working days, units per product, cooling and heating degree-days. For 2026-Q3 it learned from 2025-Q3, 2025-Q4, 2026-Q1, 2026-Q2 (362 days, R^2 0.838, out-of-sample CV(RMSE) 12.66%).

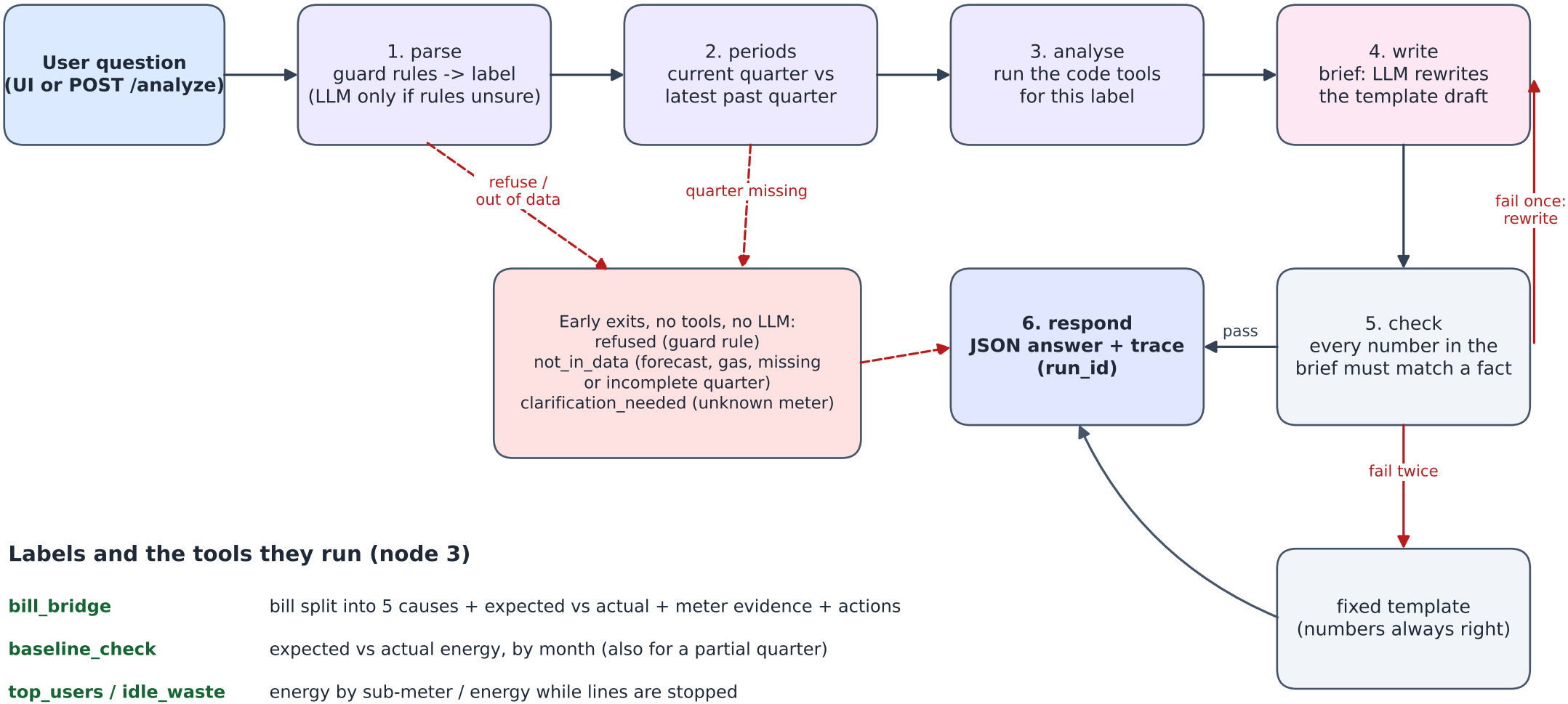




Waste slice = (actual - expected) in the current quarter minus (actual - expected) in the previous quarter. 2025-Q3 is not shown: nothing earlier exists to learn its baseline from.

Flowchart 1: what happens when a user asks a question

POST /analyze runs one LangGraph flow with 6 nodes. The LLM is used at most for labelling an unclear question and for wording the brief; code makes every number.



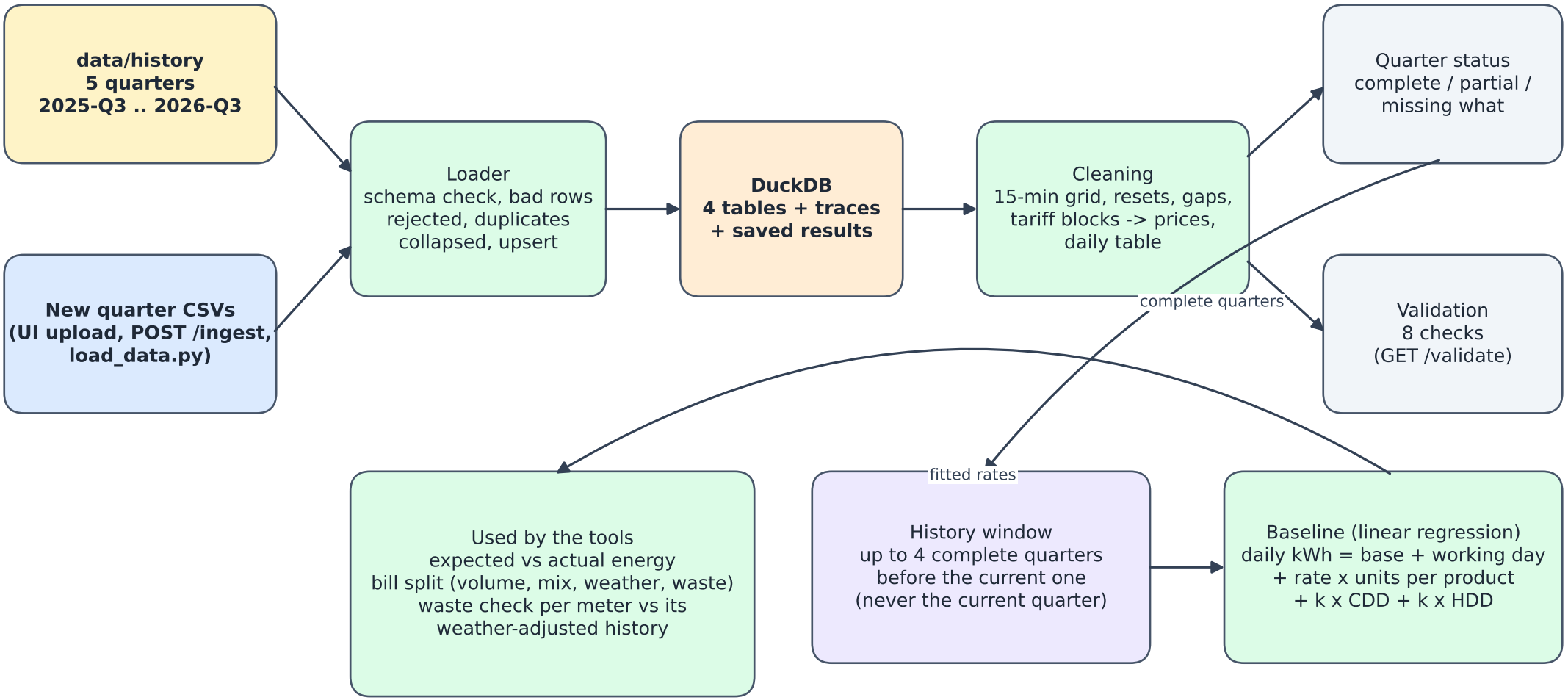
Labels and the tools they run (node 3)

bill_bridge	bill split into 5 causes + expected vs actual + meter evidence + actions
baseline_check	expected vs actual energy, by month (also for a partial quarter)
top_users / idle_waste	energy by sub-meter / energy while lines are stopped
savings / ranking	actions with kWh, money, CO2e and payback ranges
what_if	bill repriced at another tariff, energy kept as measured
bill_total / data_change	bill of a quarter / split before and after a correction

Every step (tool, inputs, outputs, timings, LLM calls, tokens) is saved in the trace: GET /trace/{run_id}.

Flowchart 2: from CSV files to answers (history + new quarter)

The history is loaded once; each new quarter is uploaded on top. A quarter is compared only when meter, production, weather and tariff cover at least 80% of its days.

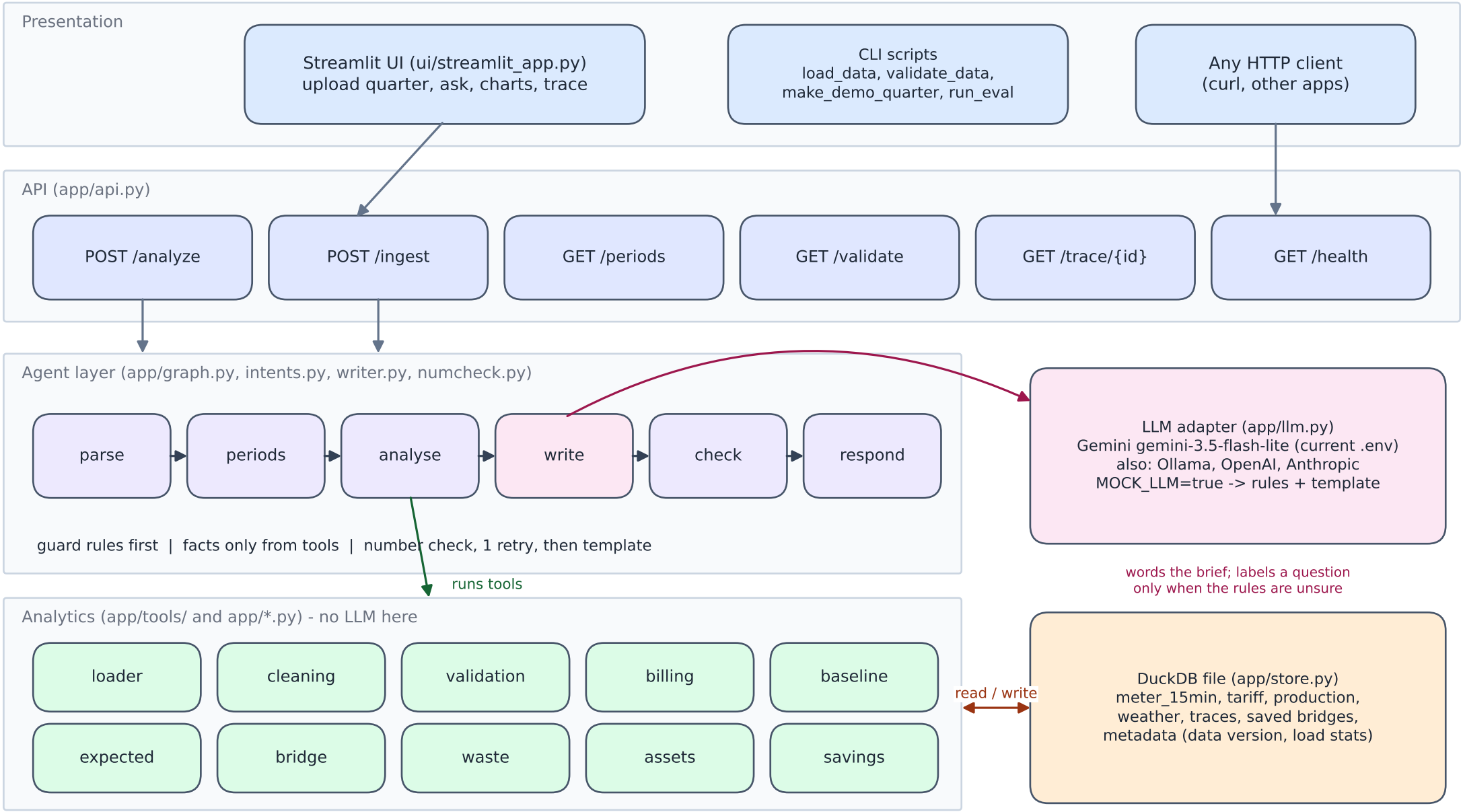


Loaded	Current	Compared with	Baseline learns from
history	2026-Q3	2026-Q2	2025-Q3, 2025-Q4, 2026-Q1, 2026-Q2
history + demo 2026-Q4	2026-Q4	2026-Q3	2025-Q4, 2026-Q1, 2026-Q2, 2026-Q3
history + 20 days of Q4	2026-Q3 (Q4 partial)	2026-Q2	same as row 1; Q4 gets expected vs actual only

Carried-forward prices never count as tariff coverage, so a forgotten tariff file keeps the quarter incomplete.

System architecture

One FastAPI service, one LangGraph flow, deterministic analytics in app/tools/, one DuckDB file. Docker Compose runs the API and the UI; start.bat does the same on Windows without Docker.



config.yaml holds every assumption (history_quarters 4, tolerance 6%, emission factor 0.4, action costs). .env holds the LLM provider, model and key (never committed).